Lie Detection via Micro Expression

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1. Motivation & Goal

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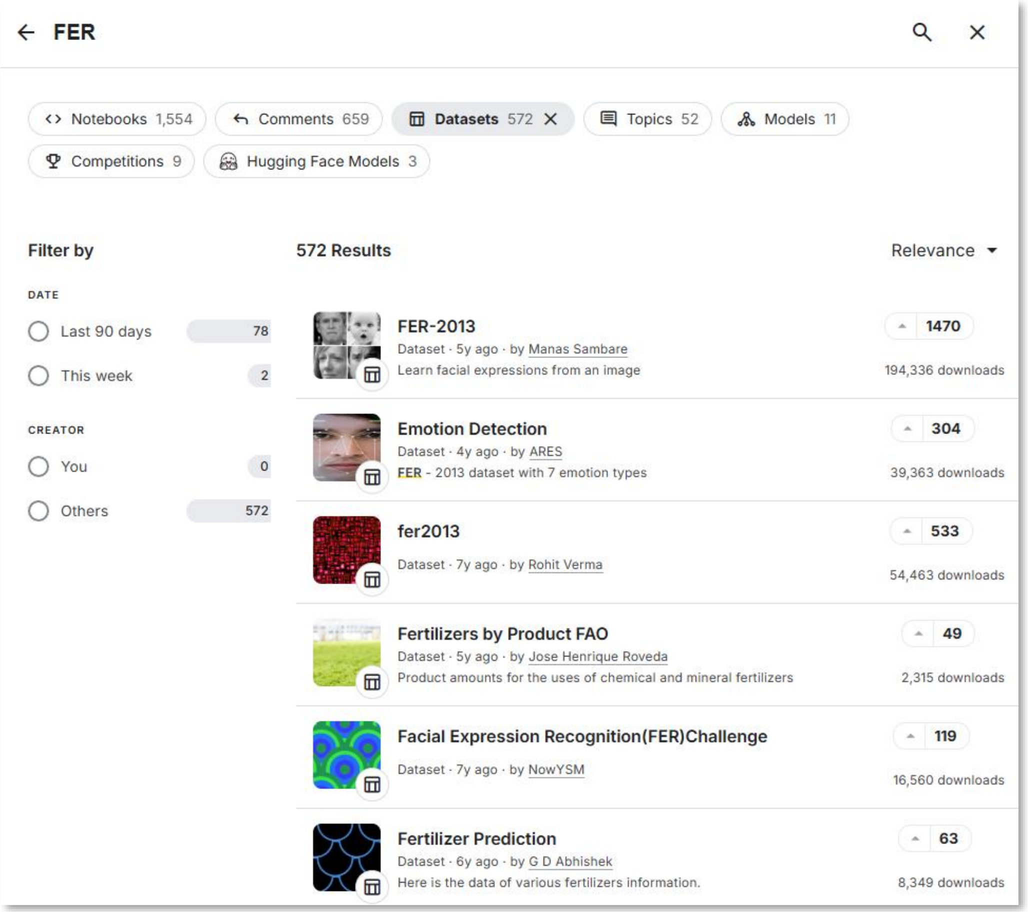
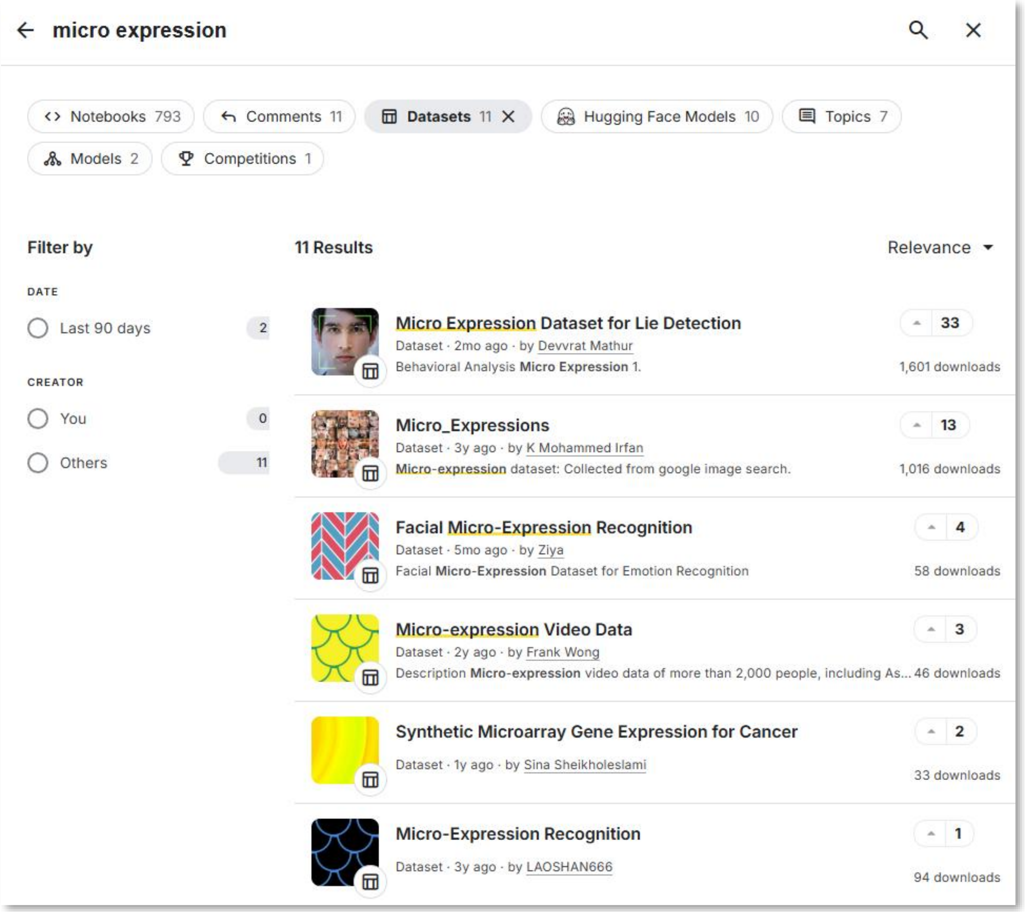
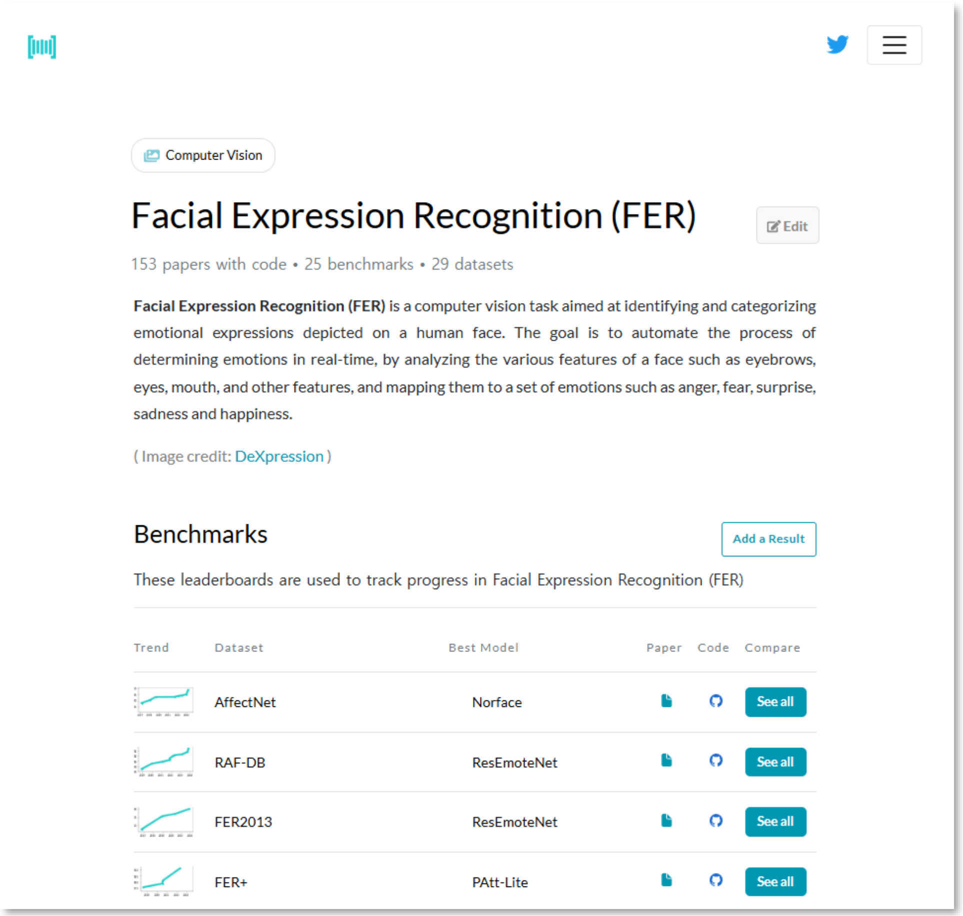


FER datasets in kaggle

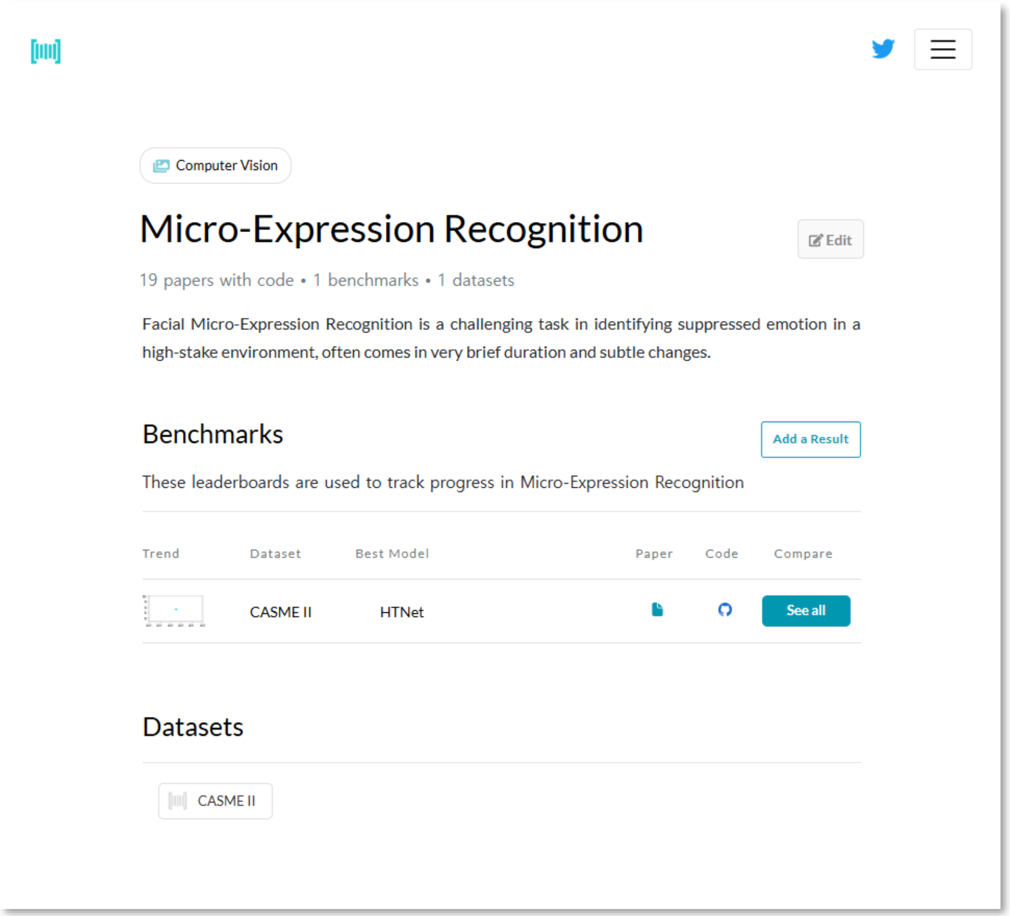


Micro-expression datasets in kaggle

1. Motivation & Goal



FER researches in paperswithcode

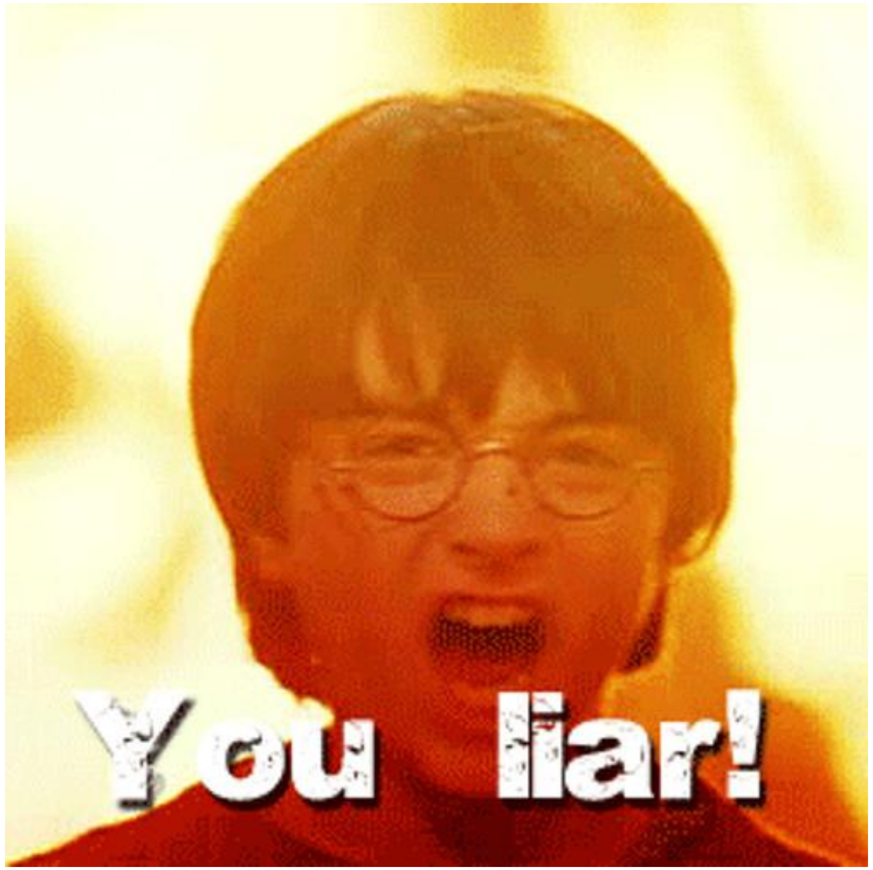


Micro-Expression researches in paperswithcode

1. Motivation & Goal

Goal

- Detect a liar by using micro expression



2. Previous Progress

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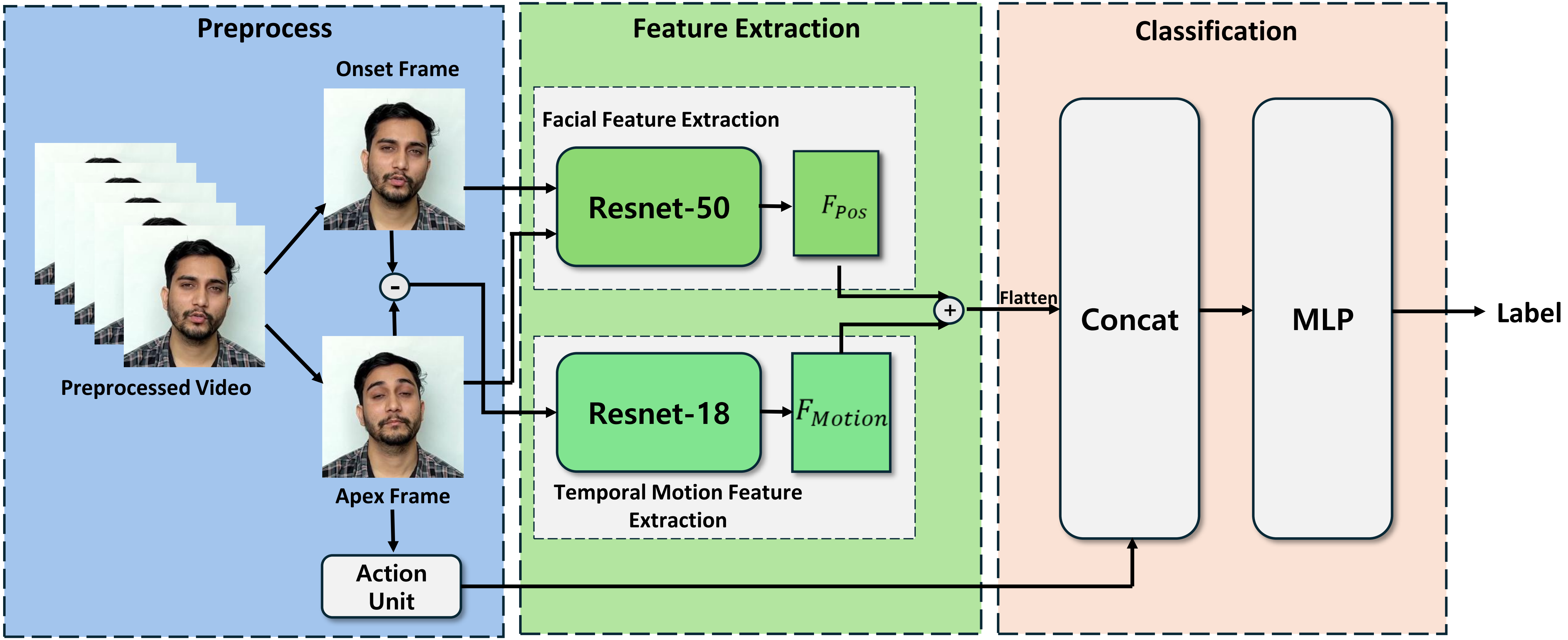

2. Previous Progress

Previous Plans

- Step: Preprocessing → Feature Extraction → Binary Classification → Evaluation
- Preprocessing: Use additional datasets to make final datasets.
- Feature Extraction : Test other methods and choose final methods.
- Binary Classification: Test other methods and choose final methods and fine tuning.
- Evaluation: Choose classification evaluation method.

3. Progress After Interim

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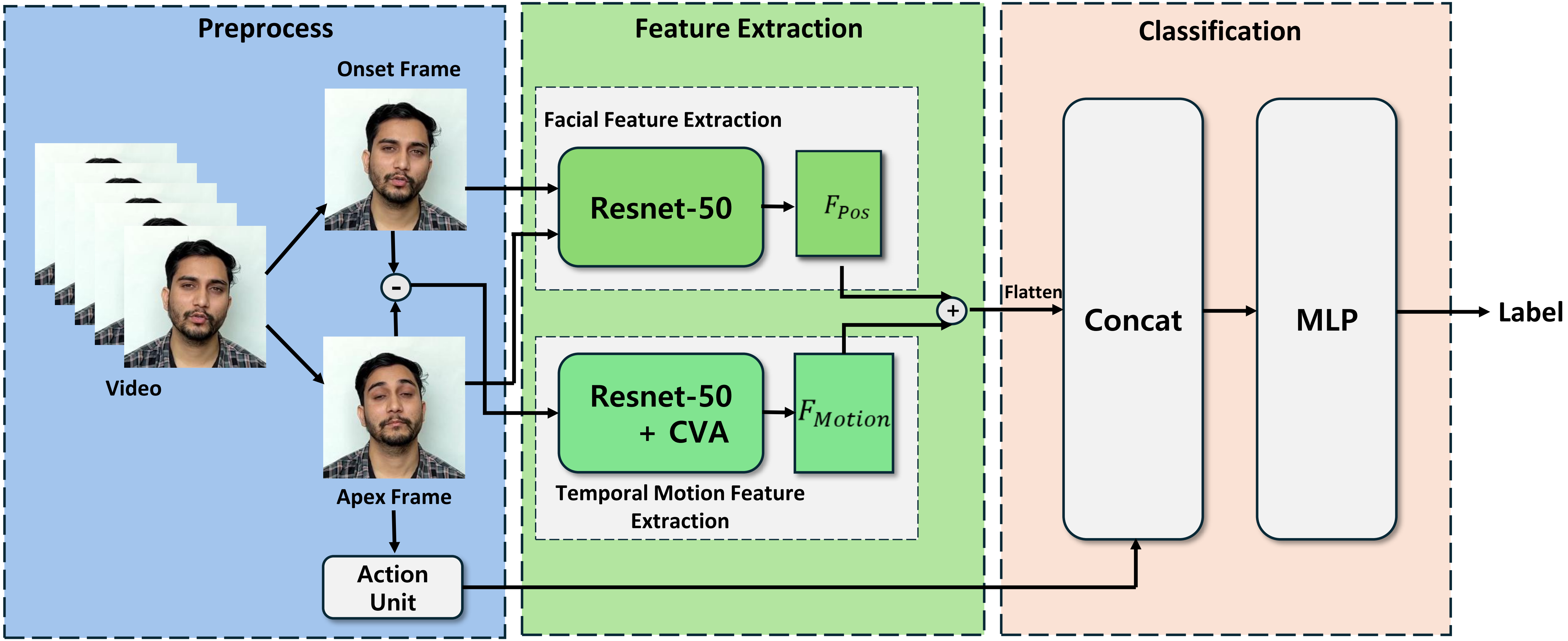


3. Progress After Interim

Data Preparation & Preprocessing

- Kaggle의 거짓말 탐지를 위한 Micro Expression 데이터셋 활용
- Real-life deception (법정 위증 데이터셋) 활용
- Bag-of-Lies 데이터셋 활용하려 했지만 실패
- 총 170개의 train data 사용

Micro Expression Dataset for Lie Detection

Detailed Facial Mirco-expression Image Dataset for Lie Detection

Real-life Deception

June 15, 2016

A multimodal dataset consisting of real-life deception: deceptive and truthful trial testimonies, manually transcribed and annotated. The dataset includes 121 short videos, along with their transcriptions and gesture annotations. (download)

Deception Detection Using Real-life Trial Data (BibTeX)

Verónica Pérez-Rosas, Mohamed Abouelenien, Rada Mihalcea, Mihai Burzo

17th ACM International Conference on Multimodal Interaction (ICMI), 2015

3. Progress After Interim

Onset/Apex Frame Extraction

- mediapipe 라이브러리 중 Face Mesh를 이용해 얼굴 랜드마크 추출
- 이전 이미지 프레임과 현재 이미지 프레임 간의 픽셀 움직임 변화량 (motion score) 계산
  - Optical flow와 landmark displacement에 0.7:0.3의 가중치 부여

```
# Combined Score
combined_score = mag_flow * 0.7 + landmark_motion * 0.3
return combined_score
```

- threshold 이상의 변화량이 나타나는 프레임을 onset(없을 시 첫번째 프레임)으로 지정
- motion score가 최댓값이 되는 프레임을 apex로 지정

Onset Frame

Apex Frame

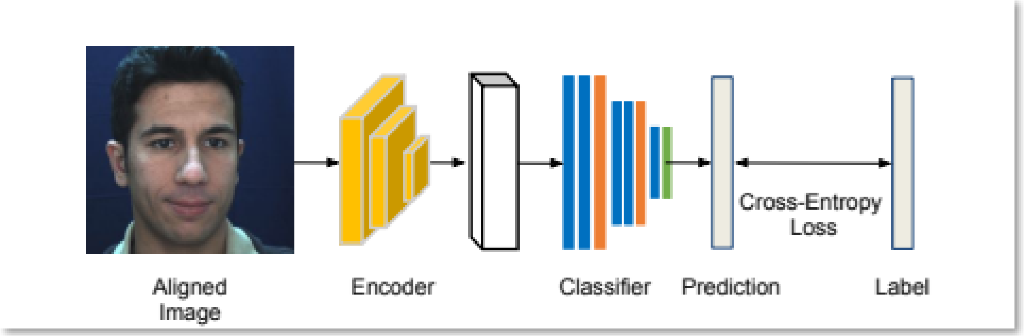
3. Progress After Interim

AU Extraction - LibreFace

- LibreFace: An Open-Source Toolkit for Deep Facial Expression Analysis
    - 얼굴 인식 → 정렬 → CNN 기반 딥러닝 모델에 적용
    - LibreFace가 인식할 수 있는 17개의 AU Unit에 대해 presence(binary)와 intensity(continuous) 예측 (Table 2.) → 총 29-dimension

Table 2. Available action units in LibreFace. **R** - LibreFace provides intensity estimation, **D** - LibreFace provides detection of occurrence.

AU	FACS name	Prediction
AU1	Inner brow raiser	R
AU2	Outer brow raiser	R
AU4	Brow lowerer	R
AU5	Upper lid raiser	R
AU6	Cheek raiser	R
AU7	Lid tightener	D
AU9	Nose wrinkler	R
AU10	Upper lip raiser	D
AU12	Lip corner puller	R
AU14	Dimpler	D
AU15	Lip corner depressor	R
AU17	Chin raiser	R
AU20	Lip stretcher	R
AU23	Lip tightener	D
AU24	Lip pressor	D
AU25	Lips part	R
AU26	Jaw drop	R



	A	B	C	D	E	F	G	H	I	J
1	au_1	au_2	au_4	au_5	au_6	au_7	au_9	au_10	au_12	au_14
2	0	0	0	0	0	0	0	1	0	0
3	au_15	au_17	au_20	au_23	au_24	au_25	au_26	au_1_inten	au_2_inten	au_4_intensity
4	0	0	0	0	0	0	0	0.107	0.055	0.563
5	au_5_inten	au_6_inten	au_9_inten	au_12_inte	au_15_inte	au_17_inte	au_20_inte	au_25_inte	au_26_intensity	
6	0.071	0.046	0.082	0.417	0.071	0.071	0.067	2.188	2.136	

3. Progress After Interim

AU Extraction - LibreFace

- 기존의 OpenFace 2.0(a facial behavior analysis toolkit) 툴보다 빠르고 정확하게 AU 판별 가능(Table 7.)
- 딥러닝 기반으로 복잡한 얼굴 움직임(micro-expression) 학습 가능
- GPU를 사용해 대량의 apex frame 이미지 처리에 적합 (Table 9.)
- 미세 표정을 기반으로 liar detection에 정량적인 AU feature 제공

Table 7. Efficiency Comparison between LibreFace and OpenFace 2.0 [4]. We report the average (Avg.) and standard deviation (Std.) of durations of each round of tests and averaged frame rate per second (FPS).

Method	Avg.↓	Std.	FPS↑
OpenFace 2.0 [4] (AU only)	50.11	0.22	19.96
LibreFace (AU only)	**25.11**	0.60	**39.82**
LibreFace (Ours)	37.20	0.66	26.88

Table 9. Efficiency comparison among CPU-only mode and GPU mode in our proposed pipeline. We report the average (Avg.) and standard deviation (Std.) of durations of each round of test, and averaged frame rate per second (FPS)

Method	Avg.↓	Std.	FPS↑
CPU-only	37.20	0.66	26.88
GPU	**6.07**	0.13	**164.82**

3. Progress After Interim

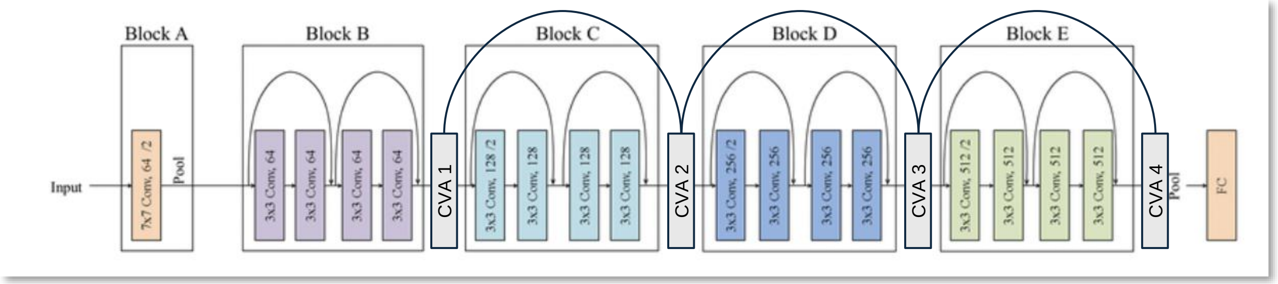
Feature Extraction - TMFE

- CNN Model: ResNet 18 Model + CVA Module
    - Vertical한 특징을 강조하여 뽑기 위해 각 layer 별 CVA Module 추가
- Onset, Apex Frame을 element wise subtraction하여 input으로 사용

Onset Frame

⊖

Apex Frame




```
tensor([[[[ 2.3398e-03,  2.2910e-03,  1.9515e-03,  ...,  1.5344e-03,
            1.6262e-03,  2.0991e-04],
          [ 0.0000e+00,  1.4738e-02,  0.0000e+00,  ...,  0.0000e+00,
            2.5973e-03,  1.2227e-02],
          [ 5.1342e-03,  3.7192e-02,  2.3743e-02,  ...,  2.0129e-02,
            1.1895e-02,  3.6625e-02],
```

• • •

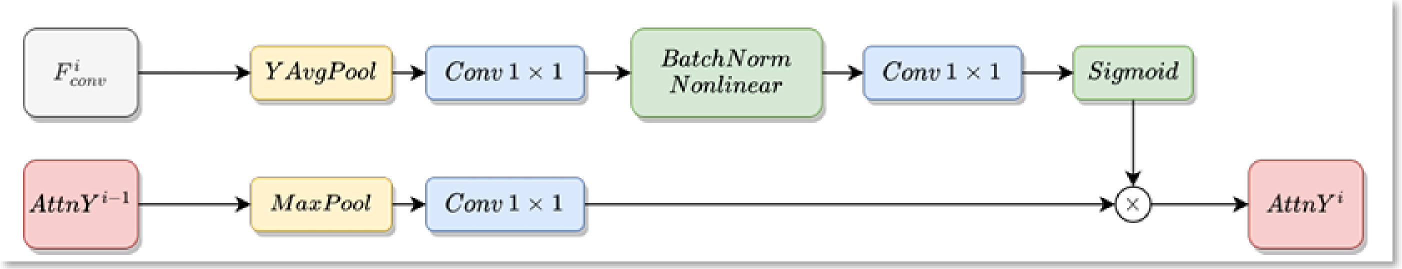
```
          [-2.5160e-02, -4.6538e-03, -3.8942e-03,  ..., -2.4790e-03,
           -1.2922e-03, -6.4192e-02],
          [-2.4224e-02,  0.0000e+00, -2.2408e-03,  ..., -4.1906e-03,
           -3.0403e-03, -6.8799e-02],
          [-4.4824e-03, -3.7549e-02, -1.0843e-02,  ..., -2.8168e-02,
           -1.0555e-02, -7.3098e-02]]]], grad_fn=<AdaptiveAvgPool2DBackward0>)
torch.Size([1, 512, 14, 14])
```

```
torch.Size([1, 512, 14, 14])
```

3. Progress After Interim

Feature Extraction - CVA Module

- CVA: Continuous Vertical Attention
    - micro expression은 vertical한 특징에 크게 영향을 받음
    - Y Avg Pool을 통해 Vertical Feature를 강화한 attention map 제작
    - 이전 attention map과 각 layer를 거친 후의 data를 합쳐 다음 attention map을 제작
- 최종 attention map과 resnet의 결과를 element-wise multiplication



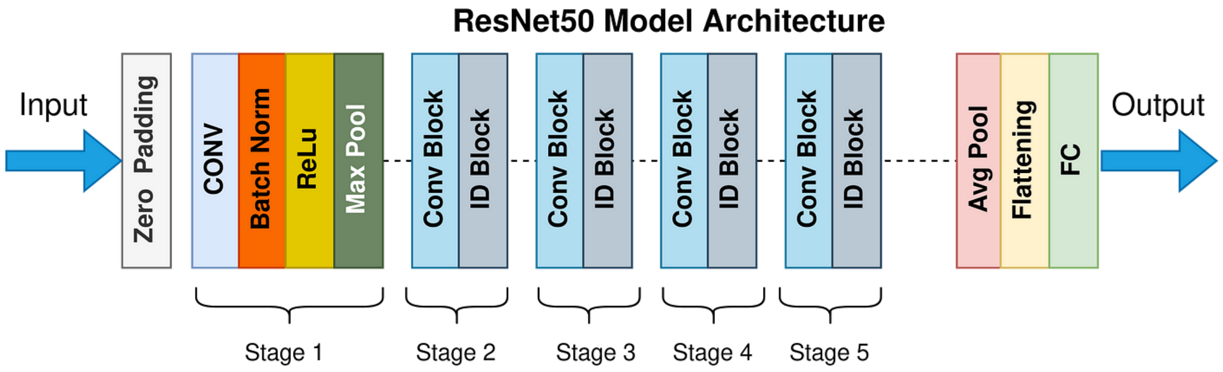
3. Progress After Interim

Feature Extraction - FPF

- CNN Model: ResNet 50 Model
- Onset, Apex Frame을 각각 넣어 output 추출 → element-wise addition
- TMFE, FPF의 결과를 concat하여 classification part로 전송

Onset Frame

Apex Frame




```
tensor([[[[0.0000, 0.0000, 0.0000,  ..., 0.0148, 0.1535, 0.0000],
          [0.0000, 0.0000, 0.0000,  ..., 0.6041, 0.7765, 0.3602],
          [0.0000, 0.0000, 0.4996,  ..., 1.6035, 2.1920, 1.6064],
          ...,
          [1.6197, 1.8867, 1.9390,  ..., 0.0000, 0.0000, 0.0000],
          [1.6312, 2.1016, 1.0926,  ..., 2.3249, 4.8814, 0.4198],
          [0.0000, 0.0000, 1.9958,  ..., 0.0000, 0.0000, 1.3007]],
```

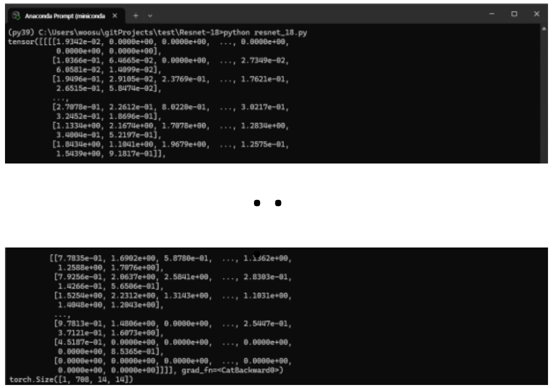
```
         [[0.9767, 0.0000, 0.0000,  ..., 0.0000, 0.0000, 0.0000],
          [1.5177, 0.0000, 0.0000,  ..., 0.0000, 0.0000, 0.0000],
          [2.5181, 0.2819, 0.0000,  ..., 0.0000, 0.0000, 0.0000],
          ...,
          [1.0867, 0.0000, 0.3025,  ..., 0.0000, 0.0000, 0.0000],
          [3.0684, 1.1337, 0.6624,  ..., 4.2551, 0.0000, 0.0000],
          [0.0000, 2.2607, 0.0000,  ..., 2.9232, 2.9928, 0.9128]]]],
       grad_fn=<AddBackward0>)
torch.Size([1, 196, 14, 14])
```

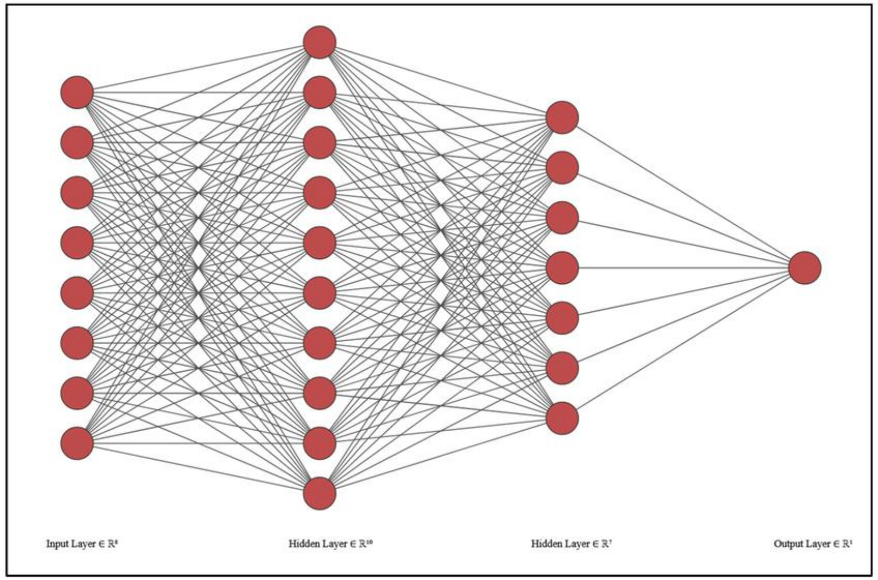
```
torch.Size([1, 196, 14, 14])
```

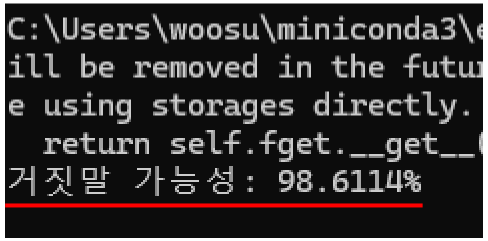
3. Progress After Interim

Binary Classification

- Binary Classification: implement binary classification model with logistic regression
- Input dim: 708 x 14 x 14 + 29 (concatenated features of FPF and TMFE + AU features)







4. Evaluation

4. Evaluation

- Binary classification 성능 평가: confusion matrix와 ROC curve 활용
    - Actual class와 prediction 비교해 TP, TN, FP, FN 계산

      → confusion matrix 생성 및 accuracy, precision, recall, f1 계산

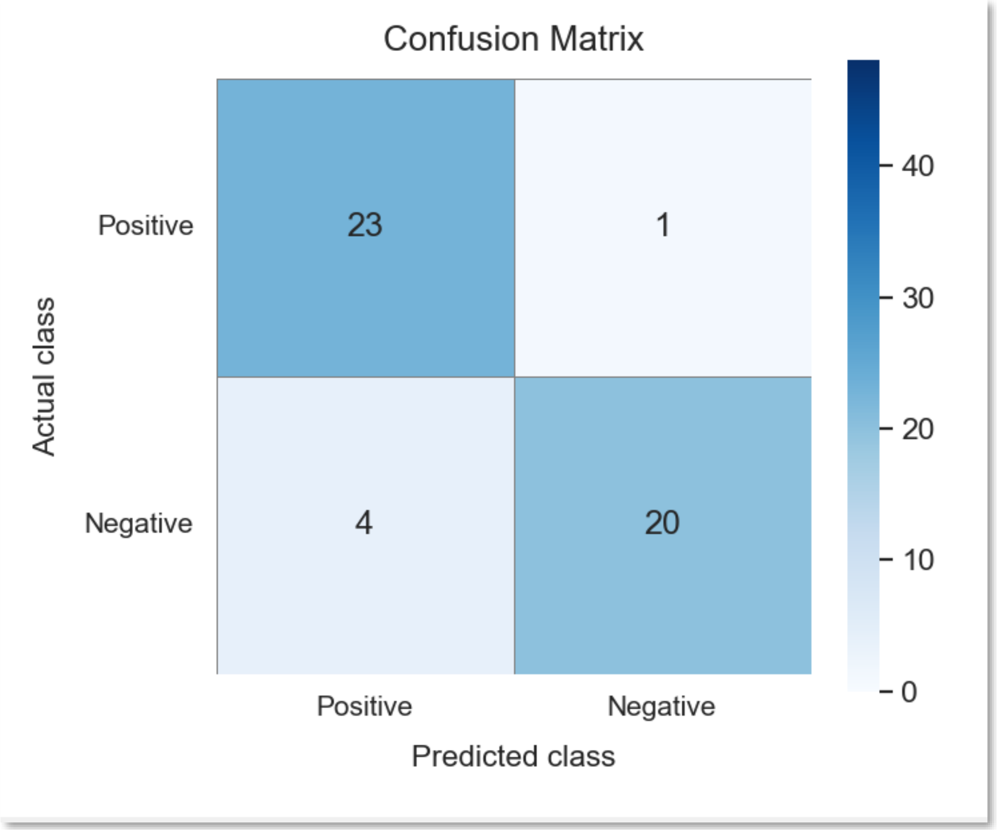
    - TPR, FPR 계산

      → ROC curve 생성 및 AUC score 계산

- 시각화: Confusion matrix, ROC curve 시각화

[1] https://en.wikipedia.org/wiki/Microexpression

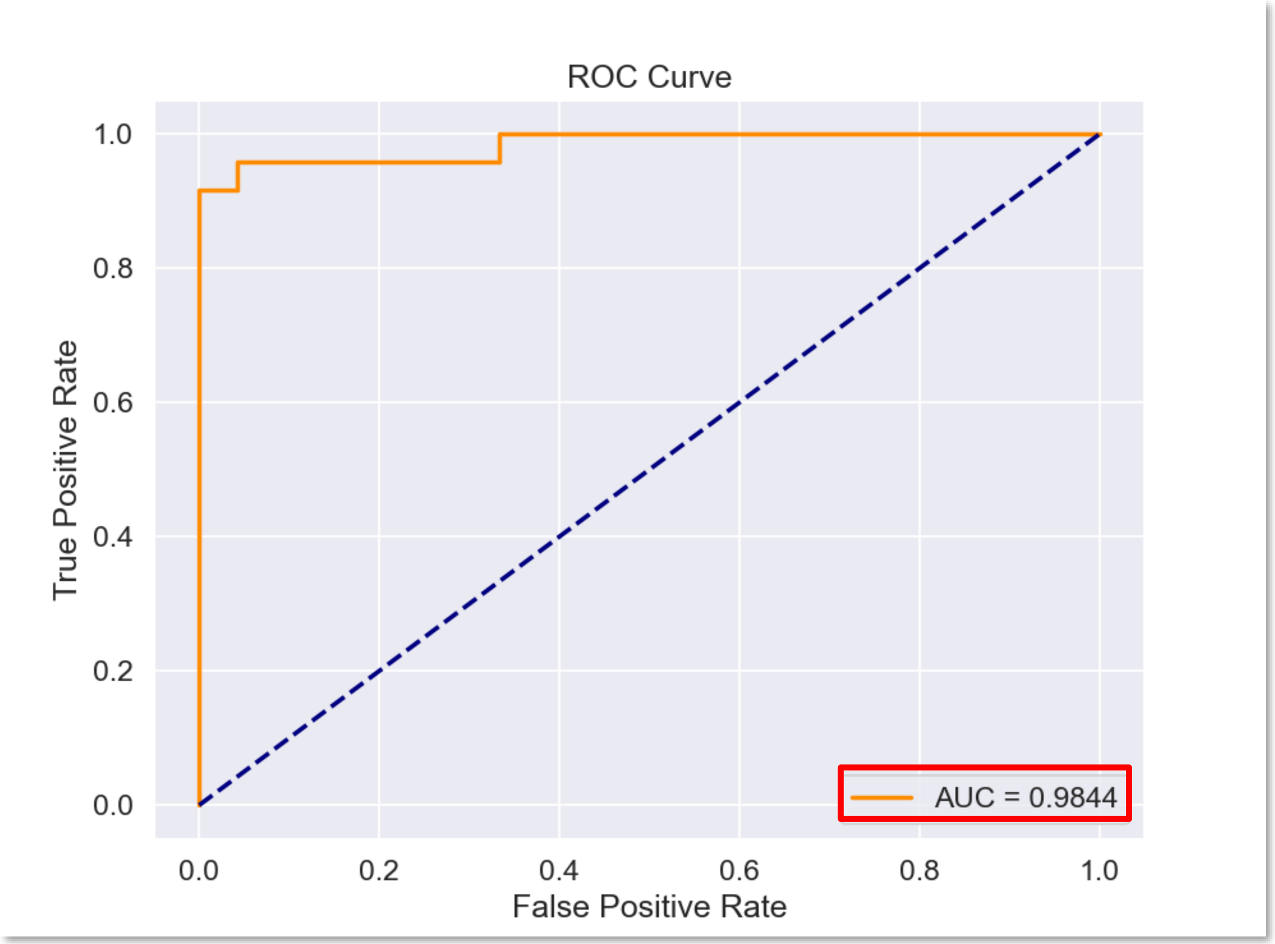
4. Evaluation



- TP: 23, FN: 1, FP: 4, TN: 20
- Accuracy = 0.8958
  - How many data were correctly predicted?
- Precision = 0.8519
  - How many data predicted as positive were correct?
- Recall = 0.9583
  - How many data of the actual positive class were correctly predicted?
- F1 score = 0.9020
  - How well did it predict the positive class in general?

[1] https://en.wikipedia.org/wiki/Microexpression

4. Evaluation



[1] https://en.wikipedia.org/wiki/Microexpression

Thanks for listening

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Reference

- https://en.wikipedia.org/wiki/Microexpression
- https://www.kaggle.com/search?q=FER+in%3Adatasets
- https://www.kaggle.com/search?q=Micro-expression+in%3Adatasets
- https://paperswithcode.com/task/facial-expression-recognition
- https://paperswithcode.com/task/micro-expression-recognition
- https://openreview.net/pdf?id=mHYkcQzdae
- https://ai.google.dev/edge/mediapipe/solutions/vision/face_landmarker?hl=ko
- https://github.com/ihp-lab/LibreFace/tree/main/AU_Detection
- https://arxiv.org/abs/2308.10713